

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Open-label Study to Evaluate the Efficacy and Safety of Sac-TMT (Sacituzumab Tirumotecan, MK-2870) Followed by Carboplatin/Paclitaxel vs Chemotherapy, Both in Combination With Pembrolizumab as Neoadjuvant Therapy for High-Risk, Early-Stage, Triple-Negative Breast Cancer or Hormone Receptor-low Positive/Human Epidermal Growth Factor Receptor-2 Negative Breast Cancer **Short Title/ Acronym:** TROFUSE-032 / MK-2870-032 **Protocol Number:** MK-2870-032 **NCT Number:** NCT06966700 **Sponsor/Company Name:** Merck Sharp & Dohme LLC / Merck **Investigational Product:** Sacituzumab tirumotecan (Sac-TMT, MK-2870), Pembrolizumab **Phase of Development:** Phase 3 **Trial Status:** Recruiting **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Director

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate if participants receiving Sac-TMT, pembrolizumab, and chemotherapy achieve a higher Pathological Complete Response (pCR) rate (fewer cancer cells found in tumors and lymph nodes during surgery) and improved Event-Free Survival (EFS) compared to those receiving only pembrolizumab and chemotherapy. **Secondary Objectives:** To evaluate Overall Survival (OS) and other pCR subsets, as well as the safety and tolerability profile of the Sac-TMT combination regimen.

2.2 Background & Rationale To address the critical need for improved therapies in high-risk, early-stage breast cancer, specifically triple-negative breast cancer (TNBC) and hormone receptor (HR)-low positive/HER2-negative breast cancer. These cancer subtypes carry a higher risk of progressing or recurring after standard treatment. Sacituzumab tirumotecan (MK-2870) is a targeted therapy and antibody-drug conjugate (ADC) designed to control how these specific types of cancer cells grow and spread.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Pembrolizumab + Chemotherapy) **Blinding:** Open-label **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target **\$\$\$:** 2,400 participants **Number of Sites:** Approximately 249-251 global locations **Estimated**

Study Start: 2024 **Estimated Primary Completion:** TBD (Long-term follow-up for Event-Free Survival)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults ≥ 18 years of age. 2. Has previously untreated high-risk, early-stage, non-metastatic (M0) breast cancer (BC), defined as any of the following combined primary tumor (T) and regional lymph node (N) staging: cT1c, N1-N2; cT2, N0-N2; cT3, N0-N2; or cT4a-d, N0-N2. 3. The participant must have a centrally confirmed diagnosis of BC that is triple-negative or HR-low+/HER2- (defined as estrogen receptor (ER)-low+ expression in 1% to 10% cells and HER2-). 4. Provides a core needle biopsy from the primary breast tumor at screening to the central laboratory. 5. Has Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1 performed within 28 days before treatment randomization. 6. Demonstrates adequate organ function.

4.2 Exclusion Criteria 1. Metastatic (Stage IV) breast cancer or clinical node stage 3 (cN3) nodal involvement. 2. Has received any prior treatment, including radiation, systemic therapy, and/or definitive surgery for currently diagnosed breast cancer. 3. Has undergone excisional biopsy of the primary tumor, axillary lymph node dissection, and/or axillary sentinel lymph node biopsy prior to study treatment. 4. Received prior systemic anticancer therapy including investigational agents within 4 weeks before randomization. 5. Received prior therapy with an anti-PD-1, anti-PD-L1, or anti-PD-L2 agent, or with an agent directed to another stimulatory or coinhibitory T-cell receptor (eg, CTLA-4, OX-40, CD137). 6. Received prior treatment with a TROP2-targeted antibody-drug conjugate (ADC) or a topoisomerase I inhibitor-containing ADC. 7. Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. 8. Known additional malignancy that is progressing or has required active treatment within the past 5 years. 9. Uncontrolled systemic disease. 10. History of (noninfectious) pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Neoadjuvant administration of Sac-TMT combined with Pembrolizumab and Chemotherapy (Carboplatin/Paclitaxel), followed by standard adjuvant therapy phases as appropriate. **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Neoadjuvant treatment phase prior to definitive surgical resection, followed by post-surgical monitoring.

5.2 Concomitant & Prohibited Therapy Permitted: Standard anti-emetics, supportive care, and medications for co-morbidities. **Prohibited:** Prior treatments noted in the exclusion criteria, including TROP2/topoisomerase I targeted ADCs, anti-PD-1/L1/L2 therapies, and live vaccines within 30 days of the first dose.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Core Needle Biopsy, CT/MRI Scans, ECG, Blood/Urine Tests
Baseline	Day 0	Randomization, First Dose
Interim	Treatment Cycles	Safety Labs, Efficacy Assessment (Imaging), Definitive Breast Surgery
End of Study	Follow-up	Long-term Event-Free Survival and Overall Survival monitoring

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Percentage of Participants with Pathological Complete Response (pCR) at the Time of Definitive Surgery:** pCR (ypT0/Tis ypN0) is defined as the absence of residual invasive cancer on hematoxylin and eosin evaluation of the complete resected breast specimen and all sampled regional lymph nodes. 2. **Event-Free Survival (EFS):** Defined as the time from randomization to disease progression that precludes surgery, local or distant recurrence, or death due to any cause, whichever occurs first.

7.2 Secondary Outcomes 1. Overall Survival (OS) 2. Percentage of Participants with pCR with No Ductal Carcinoma in Situ (pCR-no DCIS) at the Time of Definitive Surgery 3. Percentage of Participants with pCR at the Time of Definitive Surgery (High-risk, early-stage, TNBC subset)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via clinical assessments, blood tests, and ECGs. Specific focus on pneumonitis/ILD and standard chemotherapy/immunotherapy toxicities. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent External Data Monitoring Committee to regularly assess safety and efficacy parameters.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy endpoints (pCR and EFS). Safety Set for all

subjects receiving at least one dose of study medication. **Sample Size Justification:** \$N=2,400\$ powered to demonstrate statistically significant superiority in pCR and EFS. **Handling of Missing Data:** Standard oncology imputation methods for missing tumor assessments and censoring rules applied for EFS/OS analyses.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits to verify source data, assess protocol adherence, and ensure patient safety. **Audit Trail:** Validated Electronic Data Capture (EDC) system with full audit trail capabilities, eCRF locking, and rigorous data clarification processes.

11. Study Identifiers & Governance

Primary ID: MK-2870-032 **Registry Number:** NCT06966700 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC / Merck **Study Title:** A Clinical Study of Sacituzumab Tirumotecan (Sac-TMT, MK-2870) in People With Breast Cancer (MK-2870-032) **Official Regulatory Title:** A Phase 3, Randomized, Open-label Study to Evaluate the Efficacy and Safety of Sac-TMT (Sacituzumab Tirumotecan, MK-2870) Followed by Carboplatin/Paclitaxel vs Chemotherapy, Both in Combination With Pembrolizumab as Neoadjuvant Therapy for High-Risk, Early-Stage, Triple-Negative Breast Cancer or Hormone Receptor-low Positive/Human Epidermal Growth Factor Receptor-2 Negative Breast Cancer

12. Clinical Framework (Breast Cancer Specific)

Disease Areas: Breast Neoplasms, Triple Negative Breast Neoplasms, HR Low-Positive/HER2-Negative Breast Neoplasms **Primary Condition:** High-Risk, Early-Stage TNBC or HR-low+/HER2- Breast Cancer **Diagnosis Threshold:** Central laboratory confirmation via core needle biopsy **Medical Methodology:** Neoadjuvant Systemic Targeted Therapy + Immunotherapy + Chemotherapy **Optimization Target:** Pathological Complete Response (pCR) prior to definitive surgery

13. Study Procedures & Methodology

Management Pathway: Neoadjuvant administration of targeted agents and chemotherapy followed by surgical resection and adjuvant follow-up **Education Protocol (HCP):** Site Initiation Visits (SIV) and continuous training on ADC toxicities (e.g., ILD/pneumonitis

monitoring) **Education Protocol (Patient):** Informed consent process detailing the combination regimen, surgery scheduling, and required imaging/scans **Quality Audit Mechanism:** Centralized independent pathology review of surgical samples and central blinded review of radiological images

14. Observation & Follow-Up Schedule

Total Duration: Up to 5+ years for long-term EFS and OS survival follow-up **Onsite Visit Cadence:** Aligned with treatment cycles (e.g., Q3W during active systemic therapy) **Remote Monitoring Frequency:** Periodic safety follow-ups post-surgery **Checklist Review Interval:** Scheduled radiological imaging every 12 to 24 weeks during the follow-up period to monitor for disease recurrence

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				